

CHIPPEWA COUNTY AP, MI, USA

WMO: 727344

Lat: 46.250N

Lon: 84.467W

Elev: 244

StdP: 98.43

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 04869

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -23.0	(c) -20.7	(d) -29.0	(e) 0.3	(f) -21.7	(g) -25.9	(h) 0.4	(i) -19.7	(j) 11.8	(k) -5.5	(l) 10.8	(m) -5.0	(n) 2.7	(o) 80	(p) 0.630

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.8	28.3	20.6	27.2	19.8	25.9	18.7	22.2	26.7	21.0	24.9	19.9	23.9	4.2	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.0	16.1	24.9	19.2	14.4	23.1	18.5	13.8	22.3	66.1	27.0	61.9	25.4	58.0	23.6	26.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 10.8	(b) 9.2	(c) 8.3	(e) -26.7	(f) 31.6	(g) 3.7	(h) 2.3	(i) -29.4	(j) 33.2	(k) -31.5	(l) 34.6	(m) -33.6	(n) 35.8	(o) -36.2	(p) 37.5		
(a) 10.8	(b) 9.2	(c) 8.3	(e) -27.0	(f) 23.8	(g) 3.3	(h) 0.9	(i) -29.3	(j) 24.5	(k) -31.2	(l) 25.0	(m) -33.0	(n) 25.5	(o) -35.4	(p) 26.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	5.4	-8.8	-8.2	-3.4	3.5	10.9	16.2	18.7	18.1	14.3	7.5	0.9	-5.3		
	DBStd	10.99	6.42	5.91	5.94	4.76	4.62	3.88	3.26	2.96	4.21	4.54	4.98	5.67		
	HDD10.0	2622	583	508	417	201	44	3	0	0	9	107	276	473		
	HDD18.3	4824	842	742	674	444	234	85	36	41	134	337	524	731		
	CDD10.0	957	0	0	2	6	72	188	270	251	138	29	2	0		
	CDD18.3	118	0	0	0	0	4	20	47	34	12	1	0	0		
	CDH23.3	899	0	0	2	2	57	194	367	205	69	3	0	0		
	CDH26.7	169	0	0	0	0	12	40	78	28	11	0	0	0		
(14) Wind	WSAvg	3.8	3.8	4.1	4.1	4.5	4.2	3.6	3.3	3.3	3.4	3.8	3.9	3.8		
(15) Precipitation	PrecAvg	889	59	40	51	64	70	78	79	80	107	100	88	73		
	PrecMax	1209	109	74	120	131	129	152	177	125	221	183	173	151		
	PrecMin	637	22	7	6	29	20	21	31	37	19	30	32	7		
	PrecStd	138	21	19	27	28	28	29	36	25	46	40	34	32		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.9	7.3	15.9	21.3	27.9	30.2	31.1	29.0	28.0	22.8	16.0	8.9		
		MCWB	4.3	4.8	9.4	12.8	18.5	21.1	21.6	21.5	20.8	16.9	12.0	8.1		
	2%	DB	2.6	3.1	10.2	17.4	24.2	27.6	28.7	27.4	25.3	19.1	12.1	5.6		
		MCWB	1.8	1.7	6.7	9.7	15.7	19.8	21.1	20.4	18.9	15.1	9.7	4.0		
	5%	DB	1.4	2.0	7.3	13.9	22.3	26.0	27.3	26.0	23.0	17.1	9.1	3.4		
		MCWB	0.7	0.8	3.8	7.2	14.3	18.1	19.9	19.2	17.9	13.6	7.5	2.4		
	10%	DB	0.1	0.2	4.1	11.2	19.1	23.6	25.5	23.9	21.3	14.1	7.6	2.3		
		MCWB	-0.5	-0.7	1.7	6.1	12.8	16.8	18.7	18.2	17.2	11.2	6.1	1.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.3	5.4	11.4	14.0	19.9	22.7	23.8	22.9	22.2	18.6	12.5	7.9		
		MCDB	5.6	7.1	13.4	19.7	25.8	28.1	28.4	27.3	25.9	20.6	14.6	8.1		
	2%	WB	2.0	2.2	7.3	10.5	17.7	20.8	22.3	21.7	20.5	15.9	9.7	4.1		
		MCDB	2.5	3.3	9.6	15.7	22.7	25.6	27.0	25.5	23.6	19.0	11.8	5.1		
	5%	WB	0.8	0.5	4.1	8.1	15.6	19.4	21.0	20.4	19.2	13.8	7.9	2.7		
		MCDB	1.4	1.5	6.5	12.5	19.9	24.0	25.3	24.0	22.2	16.5	8.9	3.5		
	10%	WB	-0.7	-0.8	2.2	6.4	13.5	18.0	19.8	19.4	17.8	12.0	6.3	1.4		
		MCDB	-0.1	0.0	4.1	10.9	18.1	21.8	23.9	22.9	20.4	14.0	7.8	2.1		
(35) Mean Daily Temperature Range	5% DB	MDBR	7.8	8.5	9.1	9.4	11.3	11.2	10.8	10.1	9.6	8.0	6.6	6.8		
		MCDBR	6.5	8.2	11.1	14.7	15.4	13.9	12.7	11.7	11.1	11.0	9.0	7.4		
	5% WB	MCWBR	6.1	6.9	7.6	8.9	8.3	7.6	6.6	6.2	6.2	7.4	7.2	6.6		
		MCWBR	6.4	8.0	10.3	13.1	13.3	11.7	11.0	10.0	10.2	10.1	8.4	7.0		
(40) Clear-Sky Solar Irradiance	taub		0.284	0.290	0.316	0.353	0.367	0.386	0.386	0.376	0.361	0.329	0.300	0.277		
	taud		2.353	2.330	2.363	2.336	2.376	2.350	2.368	2.395	2.432	2.510	2.516	2.427		
	Ebn at Noon		816	887	912	903	899	879	874	871	853	837	801	788		
	Edh at Noon		73	92	104	117	117	121	117	110	97	78	64	62		
(44) All-Sky Solar Radiation	RadAvg		1.09	2.01	3.33	4.50	5.58	6.03	6.06	5.13	3.85	2.24	1.18	0.84		
	RadStd		0.12	0.21	0.37	0.50	0.46	0.42	0.31	0.33	0.30	0.26	0.16	0.08		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (34 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page